

# PROJECT BIBLE — BT-C9C4C5 v2.2 (Technical Reference)

**Format:** Machine-readable / AI-LLM ingestion  
**Tag:** BT-C9C4C5  
**Version:** 2.2 — Mainnet Compliant Core  
**Generated:** 2026-05-27  
**Inheritance Scope:** All future modules MUST conform to this document.

## §1 — SYSTEM IDENTITY

```
project_name: Bedrock ESG
project_type: Industrial Carbon Verification Platform
domain: bedrockesg.com
repo_tag: v2.2-mainnet-ready
stack: Fullstack Web Application (Server-Side Rendering)
db_engine: PostgreSQL (Prisma ORM)
blockchain_cluster: solana-mainnet-beta
notary_identity: "Aethexer Notary Node"
notary_node_id: "Sentinel Node-01"
notary_version: "2.2.0"
auth_protocol: "aethexer-v1"
```

## §2 — SOVEREIGN SETTLEMENT MODEL (70/20/10)

### §2.1 — Constants

```
split:
  FOUNDER_YIELD: 0.70 # Display label: "Asset Sovereign"
  STEWARDSHIP: 0.20 # Display label: "Platform Processor"
  PUBLIC_RESILIENCE: 0.10 # Display label: "Public Resilience"
  SUM_INVARIANT: 1.00 # MUST always equal 1.0 — runtime assertion enforced
```

## §2.2 — Display Labels (SETTLEMENT\_LABELS)

| Constant Key | Percentage | Display Label      | Description                                           |
|--------------|------------|--------------------|-------------------------------------------------------|
| SEVENTY      | 70%        | Asset Sovereign    | Direct liquidity for the Verified Asset Holder        |
| TWENTY       | 20%        | Platform Processor | Maintenance of Bed-rock ESG / Unicon / Freality nodes |
| TEN          | 10%        | Public Resilience  | Non-custodial routing to resilience grants            |

## §2.3 — Zero Greed Policy

IF (publicResilience / total) < 0.10 - 0.001 THEN status = HIGH\_FRICTION  
 HIGH\_FRICTION blocks settlement until ratio is restored.  
 Tolerance: ±0.1% for rounding errors only.

## §2.4 — Rules

- The 10% Public Resilience block is a **singular, unified block**.
  - **BANNED**: Hardcoded sub-splits within the 10% tier.
  - Dynamic routing by `assetClass` is permitted at the metadata level only ( `assetClassMetadata` JSON field).
  - Internal DB field names ( `founderYieldUsd` , `stewardshipUsd` , `publicResilienceUsd` ) are frozen for schema compatibility.
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## §3 — ASSET CLASS VARIANTS (A-E)

```
asset_class_variants:
  A:
    code: SOVEREIGN_SKIN
    label: "Sovereign Skin™ / Gripsy"
    characteristics: Thermodynamic + Kinetic
    description: Asphalt recycling, surface materials, circular construction inputs
  B:
    code: PROPERTY_INDUSTRIAL
    label: "Property – Industrial"
    characteristics: Commercial / Industrial Real Estate
    description: Warehouses, factories, logistics centers, commercial buildings
  C:
    code: PROPERTY_RESIDENTIAL
    label: "Property – Residential"
    characteristics: Residential Real Estate
    description: Single-family, multi-family, condominiums, rental properties
  D:
    code: PHYSICAL_COMMODITIES
    label: "Physical Commodities"
    characteristics: Fiat Currency, Commodities
    description: Currency destruction, commodity extraction, dead mass processing
  E:
    code: PRECIOUS_METALS_CUSTODIAL
    label: "Precious Metals Custodial"
    characteristics: Gold, Silver, Platinum
    description: Custodial pipeline with assay verification
```

### §3.1 — Precious Metals VAL Formula (Variant E)

$$\text{VAL} = (\text{Spot Price} \times \text{Weight} \times \text{purity\_percentage}) - \text{Refinery Discount}$$

- `purityPercentage` : 0.0–1.0 (e.g., 0.999 for 99.9%)
- `assayStatus` : UNASSAYED | PENDING | CERTIFIED
- `refineryDiscount` : USD flat discount
- `spotPriceAtIngest` : Captured at submission time
- `verifiedAssetValue` : Computed VAL stored on Submission record

### §3.2 — Routing Rule

- The `assetClassVariant` field on the Settlement model determines routing.
- Only the five codes above are valid. **BANNED**: Fabricated variant codes or categories.

## §4 — 7-TIER SENTINEL VERDICT MATRIX

### §4.1 — Composite Score Formula

$$\text{Composite} = (\text{docScore} \times 0.40) + (\text{llmScore} \times 0.35) + (\text{spatialScore} \times 0.25)$$

| Weight | Dimension             | Threshold | Method                               |
|--------|-----------------------|-----------|--------------------------------------|
| 40%    | Document Integrity    | ≥95%      | SHA-256 hash + magic byte validation |
| 35%    | LLM Parser Confidence | ≥75%      | AI extraction quality scoring        |
| 25%    | Spatial Correlation   | ≥75%      | Geolocation + metadata coherence     |

## §4.2 — Verdict Tiers

| Verdict                  | Trigger Condition             | Action                    |
|--------------------------|-------------------------------|---------------------------|
| AUTO_APPROVED            | ≥95% composite / 0 failures   | Settlement triggered      |
| PENDING_SOVEREIGN_REVIEW | 75–94% composite / ≤1 failure | Manual review queue       |
| PENDING_AUTHORIZATION    | 65–74% composite / ≤1 failure | Escrow hold               |
| SOVEREIGN_HOLD           | Manual trigger                | Perimeter flag            |
| ESCROW_REVIEW            | Condition-based               | Settlement hold           |
| PENDING_PLATFORM_REVIEW  | Structural issue              | Infrastructure validation |
| REJECTED                 | <65% composite OR ≥2 failures | No settlement             |

## §4.3 — Sentinel Identity

```

sentinel_node: "Sentinel Node-01"
sentinel_version: "1.0.0"
scoreHash_algo: SHA-256
scoreHash_input: JSON.stringify({ docScore, llmScore, spatialScore, compositeScore,
scoredAt })

```

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## §5 — SOLANA MAINNET NOTARY

### §5.1 — Configuration

```
memo_program_id: "MemoJV1t78Ds8p3D96uEn59N8MQWPrqnGoAYvGo46wR"
expected_public_key: "9tsq8qB4P9uPRSHrjXLdQJd93nuo4wEEtkR1tAJRAQ3"
rpc_env_var: SOLANA_MAINNET_RPC_URL
rpc_fallback: "https://api.mainnet-beta.solana.com"
secret_key_env_var: SOLANA_NOTARY_SECRET_KEY
secret_key_formats: ["base58", "JSON byte array"]
cluster: mainnet-beta
engine_name: "Sovereign Notary Bridge"
engine_version: "2.2.0"
```

### §5.2 — Memo Payload Schema

```
{
  "v": "2.2",
  "sys": "BEDROCK_ESG",
  "signer": "Aethexer Notary Node",
  "node": "Sentinel Node-01",
  "t": "<CARBON_AUDIT|TRADE_SETTLEMENT|HUM_FREQUENCY_SHIFT|SPATIAL_ATTESTATION|
BIOCHAR_CDR|METAL_RECOVERY>",
  "id": "<entityId>",
  "h": "<first 32 chars of dataHash>",
  "co2t": "<optional: carbonTonnes>",
  "usd": "<optional: valuationUsd>",
  "bt": "<optional: backtrace ID>",
  "ts": "<unix epoch seconds>"
}
```

### §5.3 — Production Guardrail

```
IF NODE_ENV === ProductionP:
  IF SOLANA_NOTARY_SECRET_KEY is empty/missing P LOG [PERIMETER ERROR] P HALT anchor-
  ing
  IF decoded keypair public key P EXPECTED_PUBLIC_KEY P LOG [PERIMETER ERROR] P HALT
  anchoring
```

### §5.4 — Identity Lock

- `signer` : **“Aethexer Notary Node”** — permanent. No personal names.
- `signerIdentity` : **“Aethexer Notary Node”** — permanent.
- `node` : **“Sentinel Node-01”** — permanent.
- Explorer URL format: `https://explorer.solana.com/tx/{signature}` (no cluster suffix).

## §6 — AETHEXER PROTOCOL AUTH (Trinity 3-Layer)

### §6.1 — Headers

| Header               | Purpose                         | Required   | Constraint               |
|----------------------|---------------------------------|------------|--------------------------|
| x-aethexer-key       | Protocol key (server-to-server) | One of two | Timing-safe compare      |
| x-aethexer-field-key | Field app key (mobile/edge)     | One of two | Timing-safe compare      |
| x-aethexer-ts        | Unix epoch timestamp            | Optional   | ±300s window             |
| x-aethexer-nonce     | Replay protection token         | Optional   | 16-128 chars, single-use |

### §6.2 — Environment Variables

```
AETHEXER_PROTOCOL_KEY: <server-side key>
AETHEXER_FIELD_KEY: <field app key>
```

### §6.3 — Nonce Lifecycle

- Stored in `IngestNonce` model (Prisma)
- TTL: 24 hours, auto-cleanup on each auth call
- Replay of consumed nonce → HTTP 409 `NONCE_REPLAY`

### §6.4 — Anti-Replay Window

```
TIMESTAMP_WINDOW_SECONDS = 300 # ±5 minutes
IF |server_time - request_ts| > 300 → REJECT (TIMESTAMP_EXPIRED)
```

## §7 — MAGIC BYTE VALIDATION

| MIME Type                          | Magic Bytes (Hex)    | Description         |
|------------------------------------|----------------------|---------------------|
| application/pdf                    | 25 50 44 46          | %PDF                |
| image/jpeg                         | FF D8 FF             | JPEG                |
| image/png                          | 89 50 4E 47          | .PNG                |
| application/vnd...spread-sheetml   | 50 4B 03 04          | PK (ZIP/XLSX)       |
| application/vnd...wordprocessingml | 50 4B 03 04          | PK (ZIP/DOCX)       |
| text/csv                           | (none)               | Structure-validated |
| PLY (spatial)                      | "ply\n" ASCII header | PLY 3D scan         |
| Splat (spatial)                    | ≥32 bytes            | Gaussian Splat      |

Mismatch → HTTP 422 `MAGIC_BYTE_MISMATCH` — hard reject.

## §8 — DATABASE GUARDRAILS

```
max_concurrent_connections: 25
idle_session_timeout: short
statement_timeout: 5s
idle_in_transaction_timeout: 30s
connection_model: ephemeral (no persistent session state)
```

- Reuse one Prisma client instance.
- Handle disconnects gracefully.
- Never rely on persistent session state across requests.

## §9 — SUBMISSION STATE MACHINE

DRAFT

- ☐ PENDING\_FORENSIC\_VERIFICATION
  - ☐ FORENSIC\_VERIFIED
    - ☐ [founderApproved = true]
      - ☐ SETTLEMENT\_COMPLETE

- Sovereign Approval Gate: `/api/v1/submissions/[id]/approve` — JWT protected
- Settlement fires 70/20/10 split + Solana anchor with backtrace

## §10 — EXITZ TOKEN ECOSYSTEM (Solana Mainnet)

```
tokens:
  EXITZ_BASE:
    mint: "44SrHT9Qwyz2jiiJkRF1zHKF2NFTgTC8m6ubPTb5qJBJ"
    supply: 2
    description: Base ecosystem token
  EXITZ_C:
    mint: "7ZPsJE3W5YXnnYgapzfsA5JxtK8VQwtTn5Jb23hgfzMK"
    supply: 1
    description: Classic Vehicles
    integrity_model: VIN/Odometer (no HUM)
  EXITZ_A:
    mint: "G11hBknKcBXUjc4kvAtAkwoHz5yMr7FTqEEYC96EnJn"
    supply: 1
    description: Fine Art
    integrity_model: VIN/Odometer (no HUM)
program: Token-2022
features: [on-chain metadata, forensic memo anchors]
mobile_assets: VIN/Odometer integrity (no HUM)
stationary_assets: HUM Seismic Baseline
```

## §11 — HUM LISTENER (Dynamic Baseline)

```
baseline_mw: 12.6
legacy_mw: 18.0 # deprecated
shift_threshold: 5%
sources:
  - NYSE/NASDAQ window
  - LSE window
  - CME/Eurex window
trigger: On-chain Memo anchor when ≥5% shift detected
api_routes:
  - GET /api/hum-telemetry (state)
  - POST /api/hum-telemetry (read)
```

## §12 — CARBON CALCULATION ENGINES

### §12.1 — Asphalt (Eurobitume A1-A3 LCA)

```
virgin_baseline: 60 kg CO2/tonne
rap_reduction: 0.4 kg/tonne per %RAP
mix_types: [HMA, WMA, CMA]
transport_factor: included
api: /api/calculate
external_calls: NONE (local engine only)
```

## §12.2 — Biochar CDR

```
api: /api/biochar
history: /api/biochar/history
stats: /api/biochar/stats
framework: Hedera Guardian
attestation: Dual-chain (Hedera HCS + Polygon ERC-1155)
```

## §12.3 — Metal Recovery

```
api: /api/metal
stats: /api/metal/stats
supported_metals: [STEEL, ALUMINUM, COPPER, BRASS, LEAD, ZINC, STAINLESS_STEEL, CAST_I
RON, TITANIUM, NICKEL]
emission_source: ICE Database standard factors
```

## §13 — EMISSION FACTOR DATABASE

```
sources:
- ICE Database v4.1 (25 factors)
- Ecoinvent v3.10.1 (15 factors)
total_factors: 40
biogenic_variance_correction: 60% (active for US, BR markets)
grid_mixes:
US: 0.419 kgCO2e/kWh
BR: 0.082 kgCO2e/kWh
EU: 0.253 kgCO2e/kWh
GB: 0.207 kgCO2e/kWh
api: /api/emission-factors
```

## §14 — SPATIAL HERITAGE PARSER

```
formats: [PLY ASCII, PLY Binary Little-Endian, Gaussian Splat]
pipeline: Volume → Material Mass → Carbon Estimate
material_densities:
asphalt: 2.4 t/m³
aggregate: 1.6 t/m³
concrete: 2.3 t/m³
features: [SH coefficients, opacity, scale, rotation detection]
api: /api/spatial
```

## §15 — BANNED PATTERNS

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### **banned:**

- Personal names in any codebase file or on-chain memo
- Operational monikers or nicknames in identity fields
- Hardcoded sub-splits within the 10% Public Resilience tier
- Fabricated node IDs (only "Sentinel Node-01" is valid)
- Fabricated asset class codes (only A–E above are valid)
- Fabricated routing tags or category labels
- External LLM API calls from /api/calculate (local engine only)
- Devnet cluster references (system is mainnet-only)
- Legacy signer names ("Zer0", personal identifiers)

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## §16 — ENVIRONMENT VARIABLES (Required)

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```
# Solana Notary
SOLANA_NOTARY_SECRET_KEY: <base58 or JSON array>
SOLANA_MAINNET_RPC_URL: <RPC endpoint>

# Aethexer Protocol
AETHEXER_PROTOCOL_KEY: <server key>
AETHEXER_FIELD_KEY: <field app key>

# Database
DATABASE_URL: <PostgreSQL connection string>

# Auth
NEXTAUTH_SECRET: <auto-configured>
NEXTAUTH_URL: <auto-configured per environment>
```

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## §17 — API ROUTE MAP

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| Route                         | Method           | Purpose                        |
|-------------------------------|------------------|--------------------------------|
| /api/calculate                | POST             | Asphalt carbon LCA             |
| /api/biochar                  | GET/POST         | Biochar CDR batches            |
| /api/biochar/history          | GET              | Historical batch query         |
| /api/biochar/stats            | GET              | Aggregate biochar stats        |
| /api/biochar/[id]/attestation | GET              | Blockchain attestation details |
| /api/metal                    | GET/POST         | Metal recovery calculations    |
| /api/metal/stats              | GET              | Metal recovery aggregate stats |
| /api/carbon-impact            | GET              | Unified carbon impact summary  |
| /api/stats                    | GET              | Dashboard summary metrics      |
| /api/analytics                | GET              | Time series chart data         |
| /api/projects                 | GET/POST         | Project management             |
| /api/locations                | GET/POST         | Location CRUD                  |
| /api/locations/[id]           | GET/PATCH/DELETE | Location by ID                 |
| /api/farmers                  | GET/POST         | Farmer management              |
| /api/farmers/[id]             | GET/PATCH/DELETE | Farmer by ID                   |
| /api/credits                  | GET/POST         | Farmer credit lifecycle        |
| /api/credits/stats            | GET              | Credit aggregate stats         |
| /api/notary                   | GET/POST         | Solana notary status/anchor    |
| /api/hum-telemetry            | GET/POST         | HUM listener state/read        |
| /api/emission-factors         | GET              | Emission factor lookup         |
| /api/spatial                  | GET/POST         | Spatial PLY parser             |
| /api/submissions              | GET/POST         | Business submissions           |
| /api/submissions/[id]/parse   | POST             | AI forensic parse              |

| Route                             | Method   | Purpose                                |
|-----------------------------------|----------|----------------------------------------|
| /api/submissions/[id]/settle      | POST     | Settlement execution                   |
| /api/submissions/[id]/certificate | POST     | Sovereign certificate PDF              |
| /api/upload                       | POST     | S3 presigned URL generation            |
| /api/consent                      | GET/POST | User consent management                |
| /api/consent/verify               | POST     | Consent verification                   |
| /api/consent/logs                 | GET      | Consent audit logs                     |
| /api/auth/login                   | POST     | JWT authentication                     |
| /api/auth/register                | POST     | User registration                      |
| /api/auth/agent-signup            | POST     | Agent signup with invitation           |
| /api/invitations                  | GET/POST | Invitation code management             |
| /api/agent/register-farmer        | POST     | Agent registers farmer                 |
| /api/agent/pending-counts         | GET      | Pending verification counts            |
| /api/agent/verification-queue     | GET/POST | Verification queue management          |
| /api/v1/forensic-ingest           | POST     | Unified forensic ingest                |
| /api/v1/telemetry-ingest          | POST     | Telemetry ingest (Aethexer)            |
| /api/v1/submissions/[id]/approve  | PATCH    | Sovereign approval gate                |
| /api/yards                        | GET/POST | <b>DEPRECATED</b> — use /api/locations |

## §18 — INHERITANCE DIRECTIVE

All future modules added to Bedrock ESG MUST:

1. Apply the 70/20/10 split via `UNIVERSAL_SPLIT` constants from `lib/universal-law.ts`.
2. Use `SETTLEMENT_LABELS` for all user-facing split references.
3. Route through one of the 5 defined `ASSET_CLASS_VARIANTS` (A-E). No new variants without explicit protocol amendment.

4. Pass Sentinel scoring before settlement. No bypass paths.
  5. Anchor all settlements to Solana Mainnet via `anchorToChain()` from `lib/solana-notary.ts`.
  6. Use "Aethexer Notary Node" as the signer identity. No personal names.
  7. Authenticate external ingest via `validateAethexerKey()` from `lib/aethexer-auth.ts`.
  8. Validate all uploaded file buffers via magic byte checks.
  9. Enforce Zero Greed Policy via `verifyZeroGreedPolicy()` before any settlement.
  10. Respect the DB connection ceiling of 25 concurrent connections.
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